

NIRBHAY KUMAR SINGH

Principal FPGA — Mixed-Signal — RF Systems Engineer

📍 Bengaluru, India

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Professional Summary

Principal FPGA and Mixed-Signal Systems Engineer with over 11 years of experience designing high-performance defence electronics, Direct RF Sampling architectures, AMD RFSoc platforms, FPGA systems, RF front-end hardware, and real-time radar signal processing.

Hands-on experience across complete product development from architecture, schematic design, FPGA implementation, hardware bring-up, laboratory validation, production support, customer demonstrations, and field deployment.

Extensive expertise in:

- Direct RF Sampling Architectures
- AMD RFSoc
- JESD204B/C Synchronization
- FPGA Architecture
- Mixed Signal Hardware
- RF/Microwave Systems
- High-Speed Digital Design
- Radar Signal Processing
- Technical Leadership

Core Technical Skills

FPGA	AMD RFSoc, Kintex UltraScale, MicroBlaze, RTL Design, VHDL, Verilog, Vivado, Vitis
Mixed Signal	Direct RF Sampling, RF ADC/DAC, Frequency Synthesis, Clock Distribution, JESD204B/C
High-Speed Interfaces	DDR4, PCIe, AXI4, AXI Stream, 10 Gigabit Ethernet, SPI, I2C, UART
RF	Radar Front-End Design, Phase Noise, Receiver Sensitivity, RF Measurements
Programming	MATLAB, Python, Embedded C, Tcl
Design Tools	Vivado, Vitis, MATLAB, Simulink, OrCAD Capture, Cadence Allegro
Lab Equipment	Oscilloscope, Spectrum Analyzer, Vector Network Analyzer, Signal Generator

Professional Experience

Deputy Manager – Radar Systems R&D
Bharat Electronics Limited (BEL), Bengaluru

Nov 2023 – Present

- Technical Lead for the architecture, development, integration and deployment of large-scale **160 × 16 coherent-channel AESA radar systems** for defence surveillance applications.
- Architected **Direct RF Sampling** receiver and transmitter subsystems using high-speed RF ADCs/DACs operating up to **12 GSPS**.
- Designed deterministic multi-board synchronization using **JESD204B/C**, SYSREF and precision clock distribution for coherent radar operation.
- Led FPGA architecture and RTL development for real-time radar processing including Digital Down Conversion (DDC), Pulse Compression, FFT, Doppler Processing, CFAR and Adaptive Digital Beamforming.
- Designed scalable high-speed data movement architecture using DDR4, AXI4-Stream and 10 Gigabit Ethernet.
- Performed FPGA implementation including RTL design, IP integration, timing closure, CDC analysis, floorplaning and hardware debugging.

- Led board bring-up, laboratory validation, production qualification, factory acceptance testing and field deployment.
- Collaborated closely with RF, PCB, FPGA, embedded software, mechanical and manufacturing teams throughout the complete product lifecycle.
- Developed technical documentation including System Requirements, Interface Control Documents (ICD), Design Specifications and Verification Plans.
- Supported customer demonstrations, technical reviews and engineering decision making throughout development.

Senior Hardware & Digital Systems Engineer
Bharat Electronics Limited (BEL)

2017 – 2022

- Designed FPGA architectures for radar receiver, exciter and digital beamforming subsystems.
- Developed radar signal processing algorithms including Pulse Compression, MTI, FFT, Doppler Processing and CFAR.
- Performed radar performance analysis covering receiver sensitivity, dynamic range, phase noise and link budget optimization.
- Supported hardware integration, system validation, environmental qualification and production release.
- Worked with multidisciplinary engineering teams including RF, FPGA, PCB Design, Mechanical and Manufacturing.
- Supported customer technical reviews, product demonstrations and qualification activities.

Mixed Signal & RF Design Engineer
Bharat Electronics Limited (BEL)

2015 – 2016

- Designed mixed-signal hardware for radar transceiver systems including RF receiver and exciter modules.
- Developed frequency synthesis architectures with ultra-low phase noise clock generation.
- Performed schematic capture, component selection, PCB design reviews and hardware validation.
- Conducted receiver chain analysis including gain, noise figure, phase noise and sensitivity calculations.
- Supported laboratory characterization using oscilloscopes, spectrum analyzers and RF test equipment.
- Worked closely with PCB layout engineers to optimize signal integrity and manufacturability.

Key Technical Achievements

- Led development of a **160 × 16 coherent-channel Direct RF Sampling AESA radar architecture**.
- Designed RF ADC/DAC based receiver architecture operating up to **12 GSPS**.
- Achieved deterministic synchronization across distributed radar processing units using **JESD204B/C**.
- Achieved receiver sensitivity better than **-154 dBm** through RF and digital architecture optimization.
- Delivered complete FPGA solutions from architecture through production deployment.
- Led multidisciplinary engineering teams across FPGA, RF, hardware and system integration.

Education

Master of Technology (M.Tech.)
 Communication & Signal Processing
 Indian Institute of Technology (IIT) Madras
 CGPA: **9.32 / 10**

2022 – 2024

Bachelor of Technology (B.Tech.)
 Electronics & Communication Engineering
 Bundelkhand Institute of Engineering & Technology
 Graduated with Honours

Selected Publications

- Phase Noise Effects on Micro-Doppler Extraction for Radar Target Classification Using Machine Learning
IEEE APMC 2024, Bali, Indonesia
- Advanced Deep Learning Technique for Digital Predistortion to Improve Power Amplifier Linearity
IEEE MAPCON 2024
- Multi-board Synchronization of Direct RF Digital Transceiver using JESD204 Interface
IEEE I2CT 2024

Additional Academic Contributions

- 7 IEEE Publications
- 6 Indian Patents
- Multiple International Conference Presentations

Awards & Recognition

- BEL R&D Excellence Award (Innovation)
- BEL R&D Excellence Award (Technology)
- IETE Young Scientist Award Nomination
- Best Technical Poster Award (Defence AI)

Professional Highlights

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|-----------------------------|------------------------|
| • FPGA Architecture | • PCIe |
| • Direct RF Sampling | • 10 Gigabit Ethernet |
| • Mixed Signal Design | • AXI4 / AXI Stream |
| • RF Front-End Design | • Timing Closure |
| • Radar Systems | • CDC Analysis |
| • RFSoc | • Hardware Bring-up |
| • High-Speed Digital Design | • System Integration |
| • JESD204B/C | • Production Support |
| • DDR4 | • Technical Leadership |

Tools

FPGA	Vivado, Vitis, ModelSim
PCB	OrCAD Capture, Cadence Allegro
Programming	MATLAB, Python, Embedded C, Tcl
Lab	Oscilloscope, Spectrum Analyzer, Vector Network Analyzer, Signal Generator

Open to International Relocation

Eligible for Senior FPGA, RF, Mixed-Signal, Radar Systems and Embedded Hardware Engineering Roles